PRACTICE SHEET

1. If, $\vec{a},\vec{b},\vec{c}$ are the position vectors of corners A, B, C of a parallelogram ABCD, then what is the position vector of the corner D?

(a) $\vec{a}+\vec{b}+\vec{c}$ (b) $\vec{a}+\vec{b}-\vec{c}$

(c) $\vec{a}-\vec{b}+\vec{c}$ (d) $-\vec{a}+\vec{b}+\vec{c}$

2. What is the vector in the XY-plane through origin and perpendicular to the vector $\vec{r}=a\hat{i}+b\hat{j}$ and of the same length?

(a) $-a\hat{i}-b\hat{j}$ (b) $a\hat{i}-b\hat{j}$

(c) $-a\hat{i}+b\hat{j}$ (d) $b\hat{i}-a\hat{j}$

3. Given $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$ and $\vec{b}$ is a unit vector co directional with $\vec{a}$. If m is a scalar such that $\vec{b}=m\vec{a}$, then what is the value of m?

(a) 1/5 (b) $1/\sqrt{5}$

(c) 1/29 (d) $1\sqrt{29}$

4. The magnitude of the vectors $\vec{a}$ and b are equal and the angle between them is 60° the vectors $\lambda\vec{a}+b$ and $\vec{a}-\lambda b$ are perpendicular to each other them what is the value of λ?

(a) 1 (b) 2

(c) 3 (d) 4

5. Consider the diagonals of a quadrilateral formed by the vectors $3\hat{i}+6\hat{j}-2\hat{k}$ and $4\hat{i}-\hat{j}+3\hat{k}$. The quadrilateral must be

(c) 1 (d) 1/2

11. Let $\vec{a}=(1,-2,3)$ and $\vec{b}=(3,1,2)$ be two vectors and $\vec{c}$ be a vector of length 1 and parallel to $(\vec{a}+b)$. What is $\vec{c}$ equal to?

(a) $\frac{1}{\sqrt{14}}(-2,-3,1)$ (b) $\frac{1}{\sqrt{2}}(1,0,1)$

(c) $\frac{1}{\sqrt{42}}(-5,-4,-1)$ (d) None of these

12. If $\vec{r_1}=\lambda\hat{i}+2\hat{j}+\kappa$, $\vec{r_2}=\hat{i}+(2-\lambda)\hat{j}+2\hat{k}$ are such that $|\vec{r_1}|>|\vec{r_2}|$ then λ satisfies which one of the following?

(a) λ=0 (b) λ=1

(c) λ<1 (d) λ>1

13. If P, Q, R are the mid-points of the sides AB, BC, CA respectively of a ΔABC and $\vec{a},\vec{p},\vec{q}$ are the position vectors of A,P, Q respectively, then what is the position vector of R?

(a) $2\vec{a}-(\vec{p}-\vec{q})$ (b) $(\vec{p}-\vec{q})-2\vec{a}$

(c) $\vec{a}-(\vec{p}-\vec{q})$ (d) $\frac{\vec{a}}{2}-\frac{(p-q)}{2}$

14. If $\vec{a},b$ and $\vec{c}$ are unit vectors such that $\vec{a}$ is perpendicular to the plane of $\vec{b},\vec{c}$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{}{3}$.

Then, what is $|\vec{a}+b+\vec{c}|$?

(a) 1 (b) 2